

Jakob Spurgat

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PROFESSIONAL SUMMARY

Mechanical and electrical engineer with eight years of experience developing automated test systems, engineering software, and mechanical products in the automotive industry. Excellent Python and VBA software development skills to improve engineering workflows, automate testing, and solve complex technical problems. Experienced in design for manufacturing, reverse engineering, robotics, root cause analysis, and rapid development of custom engineering tools.

TECHNICAL SKILLS

- **Programming** – Python, VBA (Excel), C++, Java
- **Vehicle Development** – mechanical mechanism design, data acquisition, test automation, OBD-II diagnostics, CAN / LIN reverse engineering
- **CAD Programs** – OnShape, NX, Inventor, Fusion360, CATIA
- **Engineering Software** – Intrepid VehicleSpy, MSC Adams, Creaform Handyscan
- **Technical Areas** – mechanical design, validation, root cause analysis, automation, robotics, data analysis, embedded systems, control systems, rapid prototyping

EXPERIENCE

Chip Ganassi Racing – Marysville, OH R&D Engineer July 2022-Present

- Provide on-site engineering support for a passenger vehicle manufacturer
- Develop tools, software, automation and processes to improve R&D and testing efficiency
- Create Excel/VBA tools that automate engineering calculations, report generation, data validation, and repetitive tasks
- Reverse engineer CAN and LIN communications on competitor vehicles so detailed tests can be performed on individual components
- Design and build a robotic system to automate legacy test equipment
- Develop a searchable database of historical test results
- Design and build a custom OBD-II interface to restore prototype vehicles from limp mode
- Create an automated weather monitoring station and integrated environmental corrections into vehicle test processing

Magna Seating – Novi, MI Product Engineer July 2018-July 2022

- Designed automotive seat structures and mechanisms while balancing safety, cost, manufacturability, and customer requirements
- Supported production launches by identifying root causes of manufacturing issues and implementing corrective actions
- Developed engineering simulations to evaluate designs before prototype builds
- Created software tools that improved engineering productivity and design processes

Magna Seating – Novi, MI Engineering Co-op June 2013-July 2018

- Conducted experiments and analyzed prototype cooled-seat performance
- Evaluated the effects of design parameters through structured testing

EDUCATION

Kettering University – Flint, MI Magna Cum Laude Graduated June 2018

- Bachelor of Science, Mechanical Engineering
- Bachelor of Science, Electrical Engineering

LEADERSHIP

First Robotics Mentor – 2017-Present

- Mentor high school students in software development, control systems, and robotics
- Helped transform a team from inconsistent regional performance into a consistent top contender, culminating in a World Championship finals appearance